Weight loss and vomiting - Gastric cancer

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Mechanical GI obstruction

3 important concepts

- Obstruction of a tubular structure – extrinsic, wall, inside lumen
- Different levels of obstruction – sequence of presentation differs
- Upper GI obstruction – relation to ampulla of Vater ➔ bile-stained or not

Mechanical GI obstruction

Upper GI obstruction

- Nature of vomitus
  - Bile stained
  - Not bile stained
- Bulge (distension) in epigastrium
- Succussion splash

- Vomiting and weight loss, gradual onset, elderly individual
  - Consider gastric cancer with outlet obstruction

Gastric adenocarcinoma

- Decreasing incidence
- Remains the 3rd leading cause of cancer deaths world-wide
- Incidence varies globally
- High incidence in Asia
- 6th leading cause of cancer deaths in HK

Epidemiology

Epidemiology

Appendix 1: Leading Cancer Sites in 2019

10 Most Common Cancers

Male

Rank	Site	No. of new cases	Relative frequency	Crude incidence rate*	Median age (yr)
1	Lung	3,424	19.4%	100.0	70
2	Colorectum	3,236	18.3%	94.5	68
3	Prostate	2,532	14.3%	74.0	70
4	Liver	1,448	8.2%	42.3	66
5	Stomach	759	4.3%	22.2	70
6	Non-Hodgkin lymphoma	589	3.3%	17.2	67
7	Nasopharynx	588	3.3%	17.2	56
8	Non-melanoma skin	564	3.2%	16.5	67
8	Kidney and other urinary organs except bladder	564	3.2%	16.5	65.5
10	Pancreas	530	3.0%	15.5	68
	All sites	**17,685**	**100%**	**516.7**	**68**

Female

Rank	Site	No. of new cases	Relative frequency	Crude incidence rate*	Median age (yr)
1	Breast	4,761	27.4%	116.6	58
2	Colorectum	2,320	13.3%	56.8	69
3	Lung	2,151	12.4%	52.7	68
4	Corpus uteri	1,198	6.9%	29.3	57
5	Thyroid	823	4.7%	20.1	52
6	Ovary & peritoneum	732	4.2%	17.9	54
7	Stomach	544	3.1%	13.3	69
8	Cervix	520	3.0%	12.7	55
9	Non-melanoma skin	517	3.0%	12.7	75
10	Non-Hodgkin lymphoma	431	2.5%	10.6	64
	All sites	**17,397**	**100%**	**425.9**	**62**

Both sexes

Rank	Site	No. of new cases	Relative frequency	Crude incidence rate*	Median age (yr)
1	Lung	5,575	15.9%	74.3	69
2	Colorectum	5,556	15.8%	74.0	68
3	Breast	4,793	13.7%	63.8	58
4	Prostate	2,532	7.2%	74.0	70
5	Liver	1,876	5.3%	25.0	67
6	Stomach	1,303	3.7%	17.4	70
7	Corpus uteri	1,198	3.4%	29.3	57
8	Non-melanoma skin	1,081	3.1%	14.4	71
9	Thyroid	1,059	3.0%	14.1	52
10	Non-Hodgkin lymphoma	1,020	2.9%	13.6	66
	All sites	**35,082**	**100%**	**467.3**	**66**

10 Major Causes of Cancer Deaths

Male

Rank	Site	No. of deaths	Relative frequency	Crude mortality rate*	Median age (yr)
1	Lung	2,651	30.7%	77.4	73
2	Colorectum	1,266	14.6%	37.0	74
3	Liver	1,129	13.1%	33.0	68
4	Prostate	445	5.1%	13.0	82
5	Stomach	393	4.5%	11.5	74
6	Pancreas	392	4.5%	11.5	70
7	Oesophagus	262	3.0%	7.7	71
8	Non-Hodgkin lymphoma	247	2.9%	7.2	74
9	Nasopharynx	230	2.7%	6.7	61
10	Bladder	174	2.0%	5.1	79
	All sites	**8,645**	**100%**	**252.6**	**72**

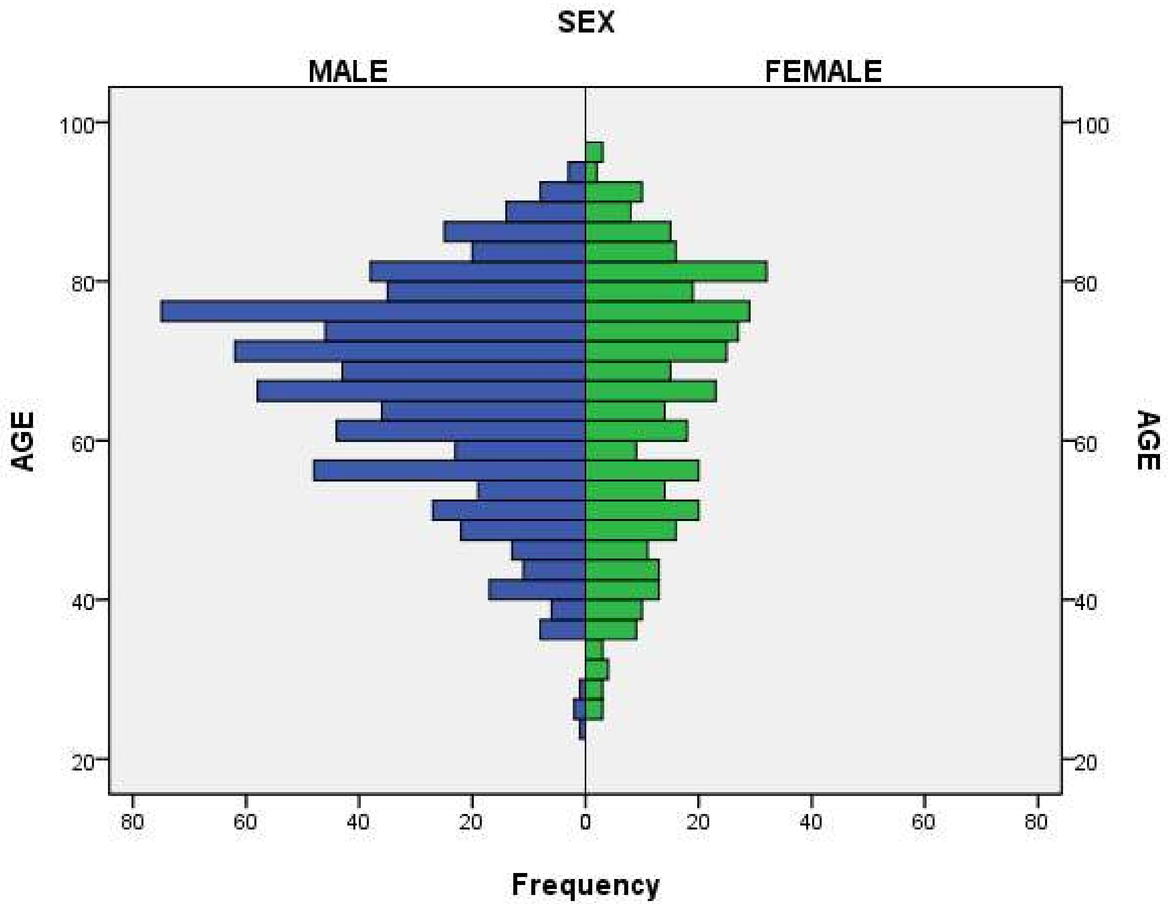
Female

Rank	Site	No. of deaths	Relative frequency	Crude mortality rate*	Median age (yr)
1	Lung	1,382	22.2%	33.8	75
2	Colorectum	908	14.6%	22.2	78
3	Breast	852	13.7%	20.9	63
4	Liver	401	6.4%	9.8	80
5	Pancreas	348	5.6%	8.5	74
6	Stomach	303	4.9%	7.4	71
7	Ovary & peritoneum	261	4.2%	6.4	62
8	Cervix	162	2.6%	4.0	62
9	Non-Hodgkin lymphoma	156	2.5%	3.8	76
10	Leukaemia	140	2.2%	3.4	70
	All sites	**6,226**	**100%**	**152.4**	**72**

Both sexes

Rank	Site	No. of deaths	Relative frequency	Crude mortality rate*	Median age (yr)
1	Lung	4,033	27.1%	53.7	73
2	Colorectum	2,174	14.6%	29.0	75
3	Liver	1,530	10.3%	20.4	70
4	Breast	859	5.8%	11.4	63
5	Pancreas	740	5.0%	9.9	72
6	Stomach	696	4.7%	9.3	73
7	Prostate	445	3.0%	13.0	82
8	Non-Hodgkin lymphoma	403	2.7%	5.4	75
9	Oesophagus	320	2.2%	4.3	72
10	Leukaemia	298	2.0%	4.0	71
	All sites	**14,871**	**100%**	**198.1**	**72**

Age distribution(QMH)



Risk factors

- Harmful dietary factors
  - N-nitroso compounds
  - preserved, smoked, salted food
- Protective dietary factors
  - trace elements
  - vitamin C
  - fresh fruits and vegetables

Risk factors

- Atrophic gastritis
- Pernicious anemia
- Adenomatous polyps
- Menetrier's disease
- Previous partial gastrectomy (>20yrs)
- Smoking (11%)
- EBV
- Industrial (dusty, high temperature, rubber, coal mining, metal processing, chromium production)
- Common variable immunodeficiency (CVID)
- Hereditary – E-cadherin mutation

Risk factors

- *Helicobacter pylori* (WHO Group 1 carcinogen)

Linitis plastica

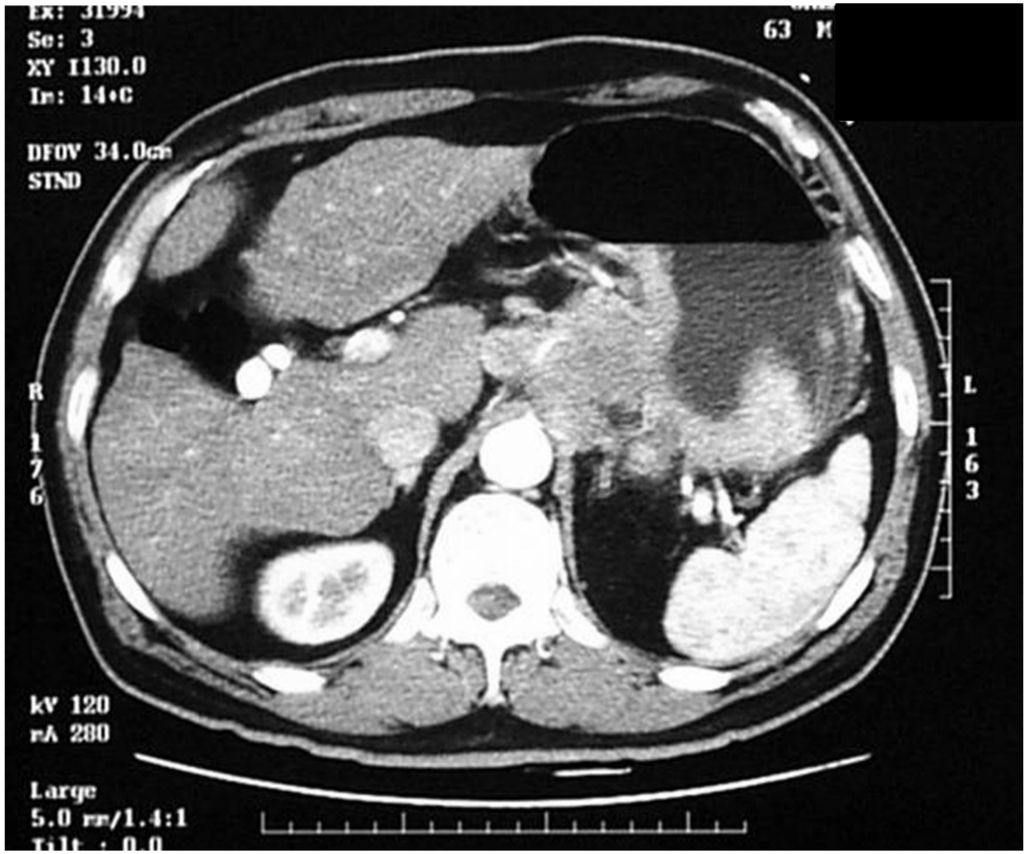
- ‘Leather bottle’ stomach
- Mucosa may appear “normal” on endoscopy
- Rigid, could not be distended with air

Mode of spread

- Direct invasion
- Lymphatic
- Transcoelomic (seedlings, Krukenberg tumour)
- Haematogenous (e.g. liver, lung)

Direct invasion

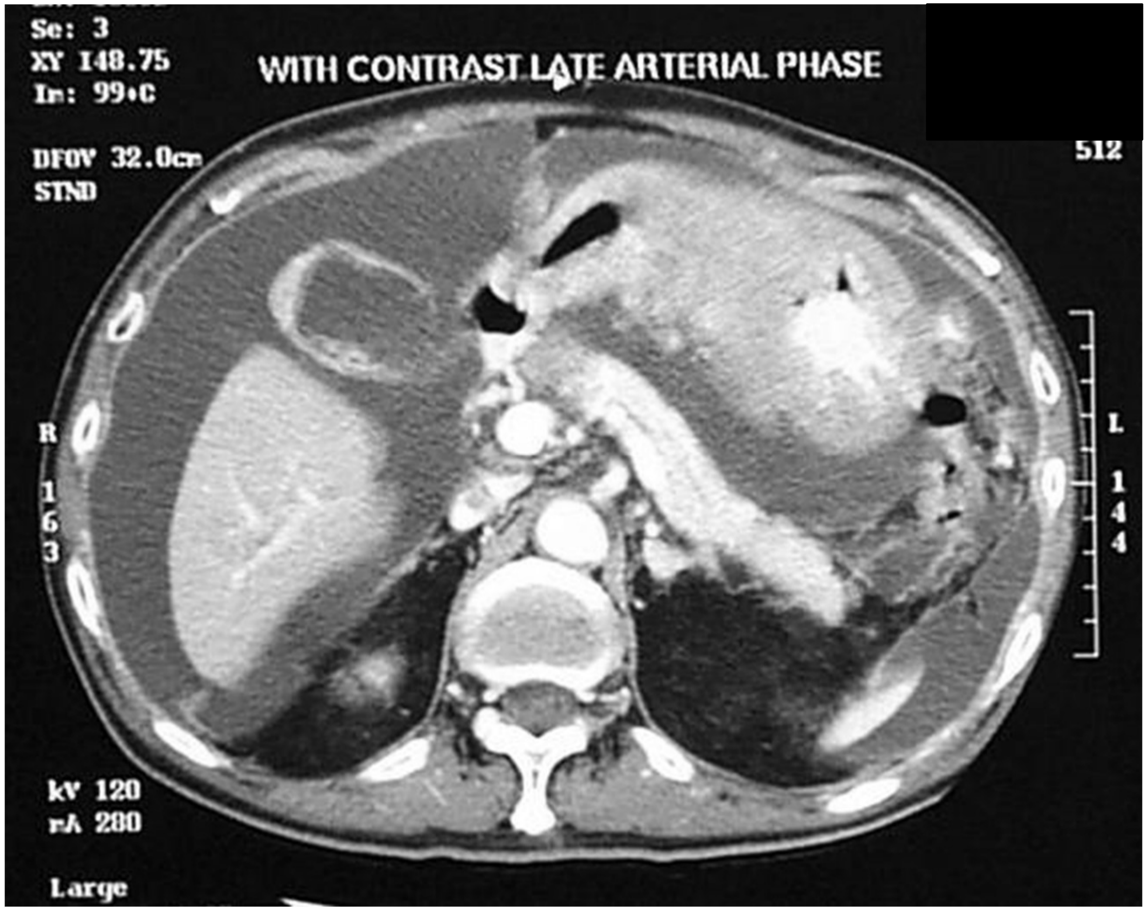
Lymphatic



Se: 3
XY I130.0
Im: 14+C
DFOV 34.0cm
STND
63 M
R
176
L
163
kV 120
mA 280
Large
5.0 mm/1.4:1

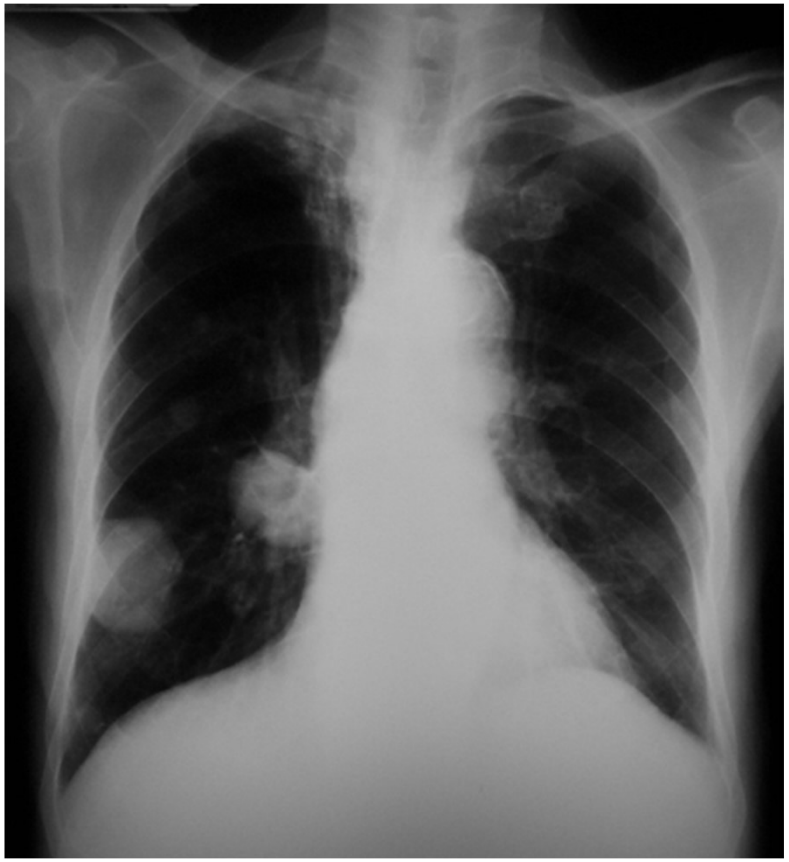
Transcoelomic

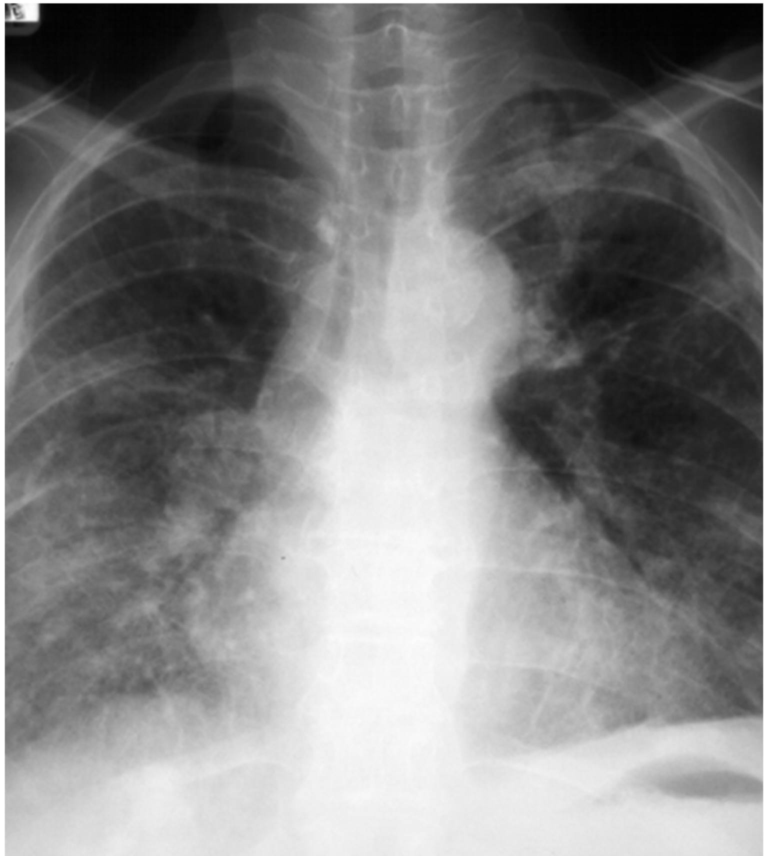
Transcoelomic



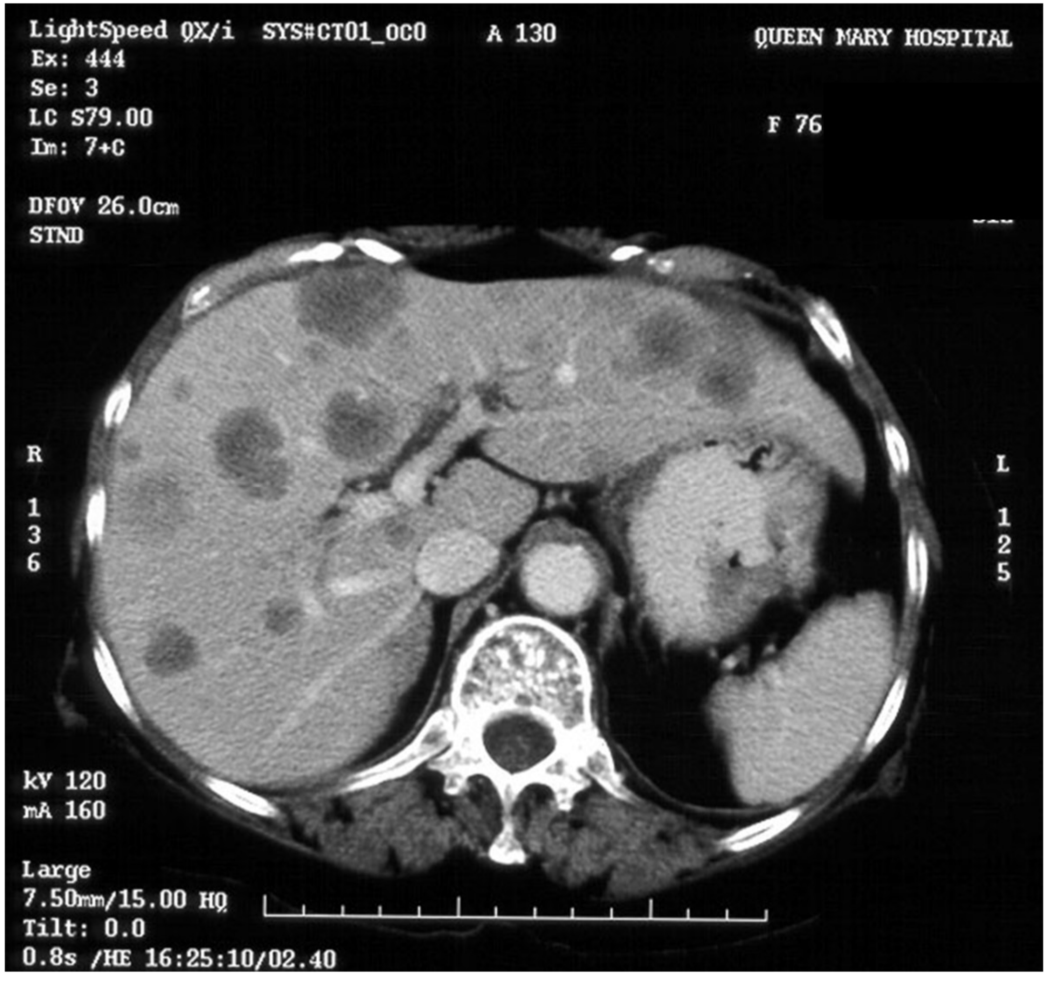
Krukenberg tumour

Haematogenous





Haematogenous



4 Molecular subtypes

EBV (8.8%)

- Prevalence in males
- Frequently located at fundus and body
- EBV-CIMP
- CDKN2A silencing
- JAK2, CD274, PDCD1LG2 and ERBB2 amplification
- PIK3CA mutation (80% subtype) inactivating in the kinase domain (exon 20)
- ARID1A (55%) and BCOR (23%) mutations
- Inmune cell signaling enrichment

MSI (21.7%)

- Prevalence in females
- An older age at diagnosis
- Gastric –CIMP
- MLH1 silencing
- Lacks targetable amplifications
- Hypermuted tumors: recurrent substitution mutations in TP53, KRAS, ARID1A, PIK3CA, ERBB3, PTEN, CHRD, HLA-B) and insertions/deletions/mutations in RNF43, B2M and NF1 genes.
- Activation of mitotic pathways

GS (19.7%)

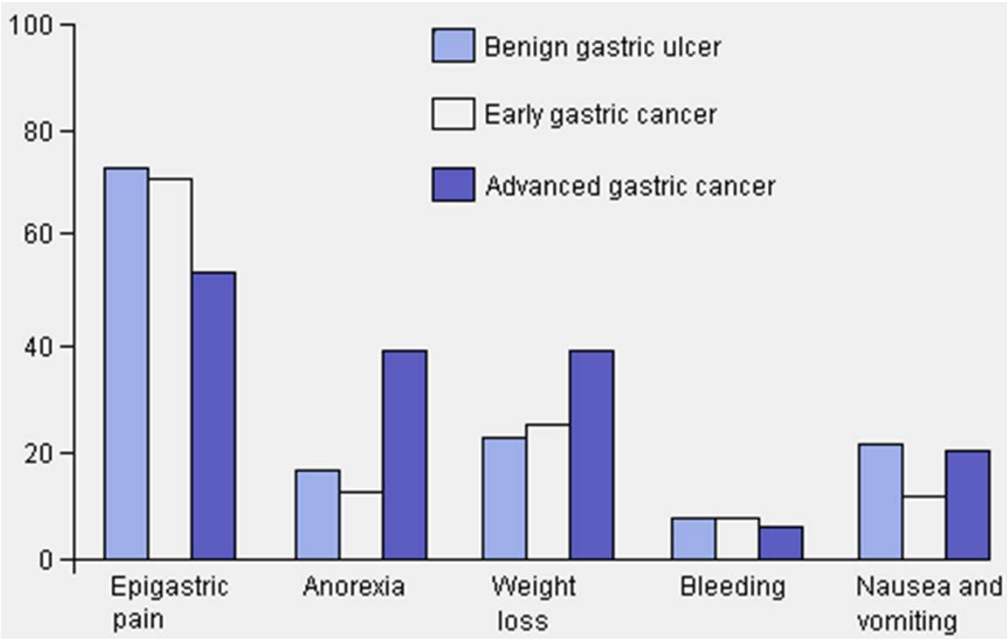
- Diffuse histology enrichment
- An early age at diagnosis
- Recurrent CDH1 (37%), RHOA mutation (15%), inactivating ARID1A mutation
- Enrichment of CLDN18-ARHGAP fusion is mutually exclusive with RHO mutations
- Signaling pathway enrichment: Cell adhesion (including integrin and syndecan-1 mediated signaling), angiogenesis

CIN (49.8%)

- Intestinal histology enrichment
- Frequently located at gastroesophageal junction/cardia
- RTK-RAS amplifications (EGFR, ERRB2, ERRB3, VEGFA, FGFR2, MET, NRAS/KRAS, JAK2, CD274, PDCD1LG2 and PIK3CA)
- Phosphorylation of EGFR (pY1068) consistent with amplification
- TP53 mutations (71%)

Clinical presentation

- Notoriously difficult to make an early diagnosis



Clinical features

- Could be asymptomatic
- Distending discomfort, vomiting (splash)
- Anaemia, pallor, melaena, haematemesis
- Perforation with acute peritonitis (rare)
- Epigastric pain

Clinical features

- Anorexia, wt loss, malaise, weakness
- Dysphagia (cardia)
- Abdominal mass (primary, omental, Krukenberg)
- Acanthosis nigricans
- Paraneoplastic syndrome – e.g. nephrotic

Clinical features

- metastatic disease
  - abdominal distension (ascites)
  - jaundice (biliary obstruction, liver metastases)
  - supraclavicular lymph node (Virchow's node, Troisier's sign), left axillary node (Irish's)
  - dyspnoea (pleural effusion, lymphangitis)
  - hepatomegaly
  - Sister Joseph's nodule
  - acute renal failure / hydronephrosis
  - rectal (Blumer's) shelf

Investigations

- CBP, LFT, RFT
- Upper endoscopy and biopsies
- Chest x-ray
- (CEA, CA19-9) (follow-up)

2 basic questions after diagnosis

- Stage of the disease
- Is the patient fit for surgery / treatment?

Staging

- TNM
- American Joint Committee on Cancer (AJCC)
- International Union Against Cancer (UICC)
- Japanese Classification of Gastric Cancer (JGCA)

TNM Staging

- T staging
  - Depth of invasion
- N staging
  - Number of lymph node(s) with metastasis
- M staging
  - Presence/absence of systemic metastasis

Clinical staging

- History and physical examination
- Liver function test
- Chest x-ray
- Ultrasonography or CAT scan abdomen
- PET/CT scan
- Endoscopic ultrasound
- Laparoscopy

Endoscopic ultrasound

Laparoscopy

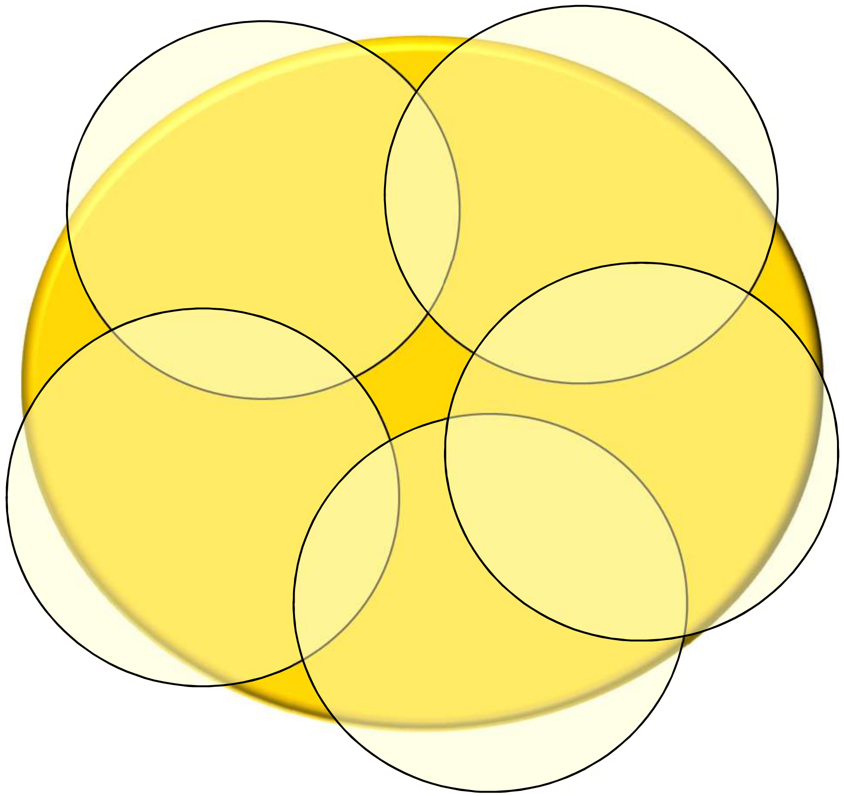
Treatment

- Depends on “fitness” and clinical stage
- Currently, resection remains the only hope for cure for resectable disease
- In HK, ~70% patients present with diseases $\geq$ Stage III

Treatment

- Early cancer
  - T1, mucosal
  - Rare in Hong Kong
  - Japan – screening endoscopy
  - Endoscopic mucosal resection (EMR)
  - Endoscopic submucosal dissection (ESD)
  - Laparoscopic gastrectomy

Endoscopic mucosal resection (EMR)



Endoscopic submucosal dissection

Resectable gastric cancer

- Gastric resection with D2 lymph node dissection
- Distal resection for distal lesion
- Total gastrectomy for proximal lesion
- Postop adjuvant chemotherapy – for advanced cancer
- Preop neoadjuvant chemotherapy – selected patients

Resectable gastric cancer

Resectable gastric cancer

Roux-en-Y
esophagojejunostomy

Gastric cancer

Laparoscopic gastrectomy

Laparoscopic reconstruction

- Laparoscopic assisted vs total laparoscopic
- Hand sewn vs stapled

Totally laparoscopic reconstruction

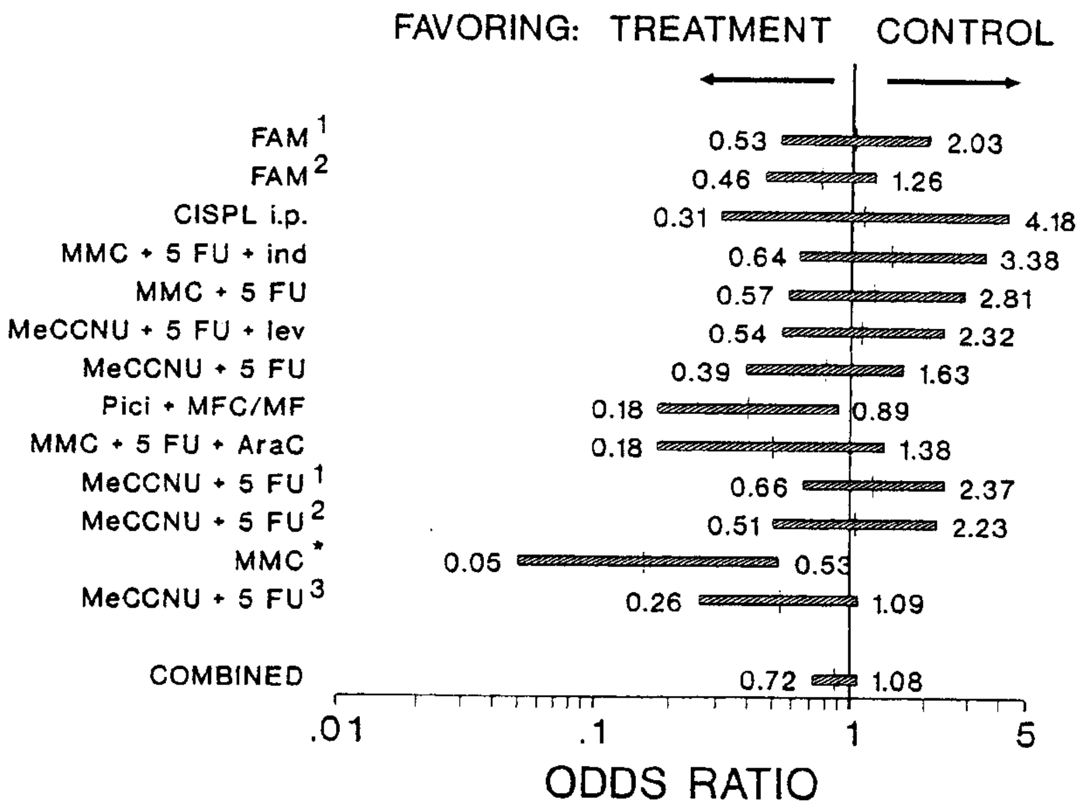
- Intra-corporeal hand-sewn anastomosis

Intracorporeal hand-sewn anastomosis

Intracorporeal hand-sewn anastomosis

Robotic gastrectomy

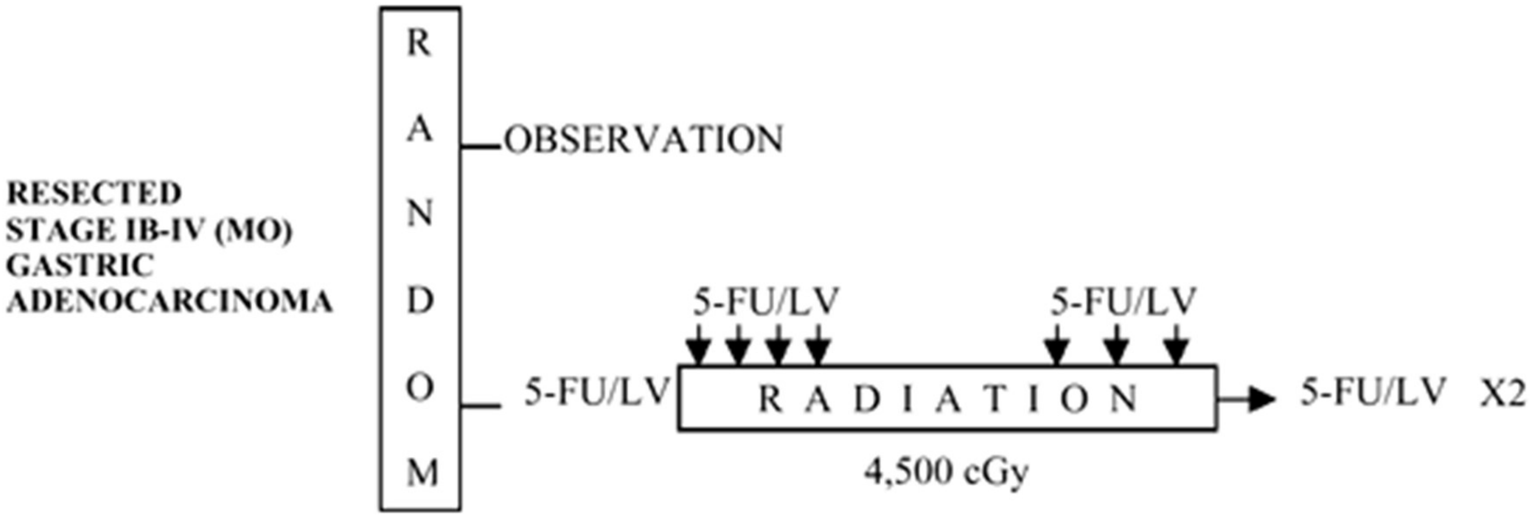
Adjuvant chemotherapy



Hermans, J, et al, 1993

Adjuvant chemoradiation

- US Intergroup 0116



Macdonald JS, *et al*, N Engl J Med 2001

Adjuvant chemoradiation

- Survival benefit
- Inadequate LN dissection (10% D2, 36% D1, 54% D0) (RT replacing LN dissection?)
- Very high morbidity (41% grade 3, 32% grade 4 toxicity)
- 1% toxic mortality

Macdonald JS, *et al*, N Engl J Med 2001

JOURNAL OF CLINICAL ONCOLOGY

ORIGINAL REPORT

Five-Year Outcomes of a Randomized Phase III Trial Comparing Adjuvant Chemotherapy With S-1 Versus Surgery Alone in Stage II or III Gastric Cancer

Mitsuru Sasako, Shinichi Sakuramoto, Hitoshi Katai, Taira Kinoshita, Hiroshi Furukawa, Toshiharu Yamaguchi, Atsushi Nashimoto, Masashi Fujii, Toshifusa Nakajima, and Yasuo Ohashi

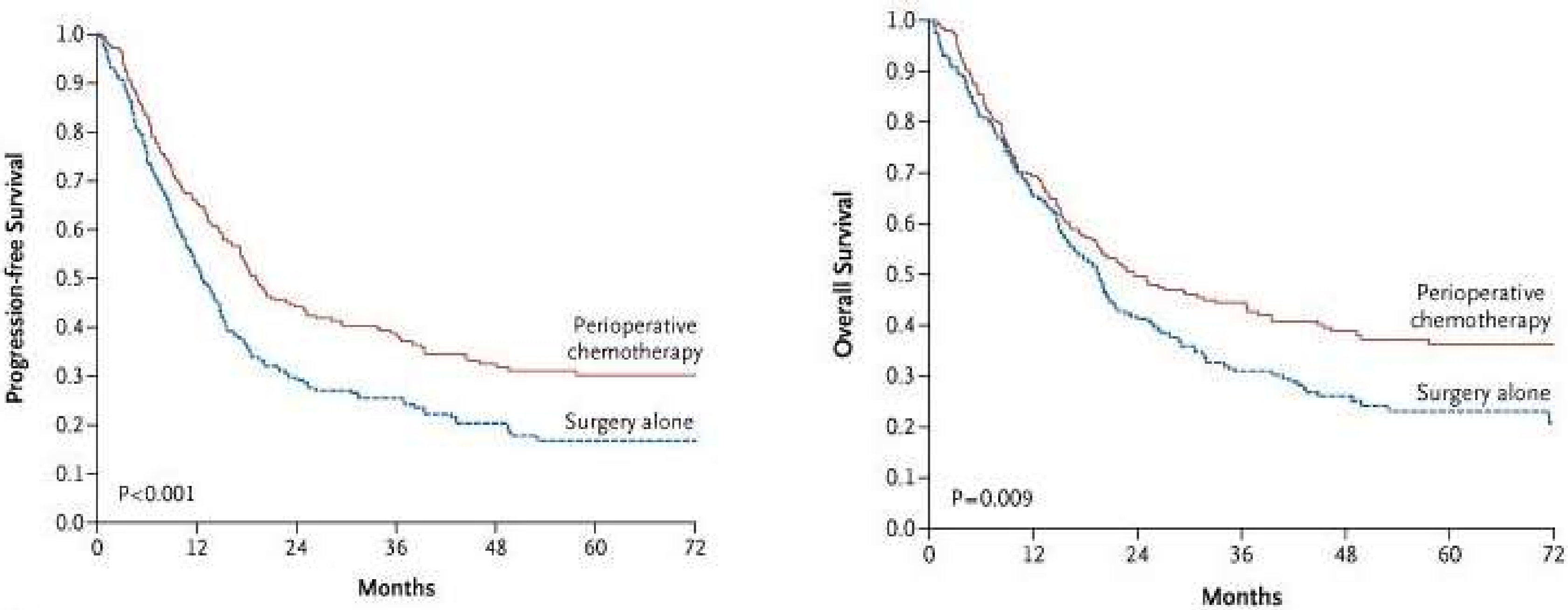
Sasako M, *et al*, J Clin Oncol 2011

Adjuvant capecitabine and oxaliplatin for gastric cancer after D2 gastrectomy (CLASSIC): a phase 3 open-label, randomised controlled trial

Yung-Jue Bang*, Young-Woo Kim, Han-Kwang Yang, Hyun Cheol Chung, Young-Kyu Park, Kyung Hee Lee, Keun-Wook Lee, Yong Ho Kim, Sang-Ik Noh, Jae Yong Cho, Young Jae Mok, Yeul Hong Kim, Jiafu Ji, Ta-Sen Yeh, Peter Button, Florin Sirzén, Sung Hoon Noh*, for the CLASSIC trial investigators†

Bang YJ, *et al*, Lancet 2012

Neoadjuvant therapy



Cunningham D, *et al*, N Eng J Med 2006

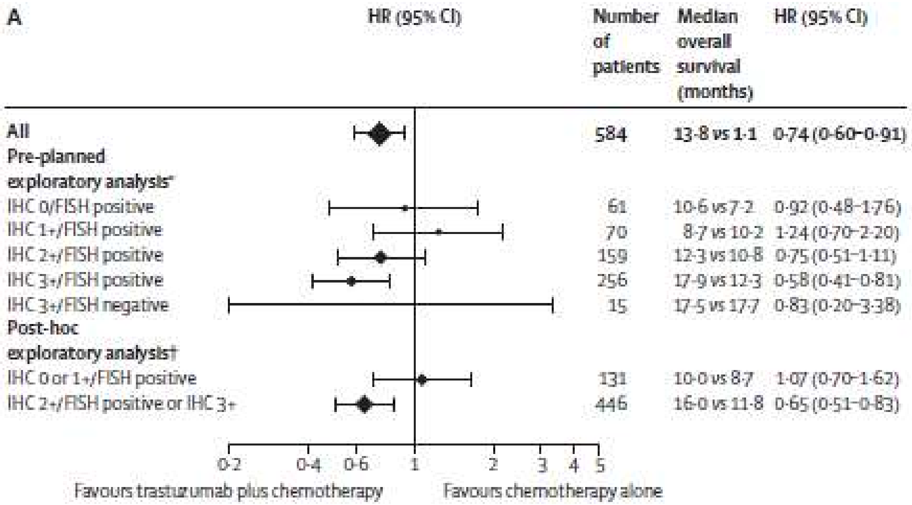
Unresectable disease

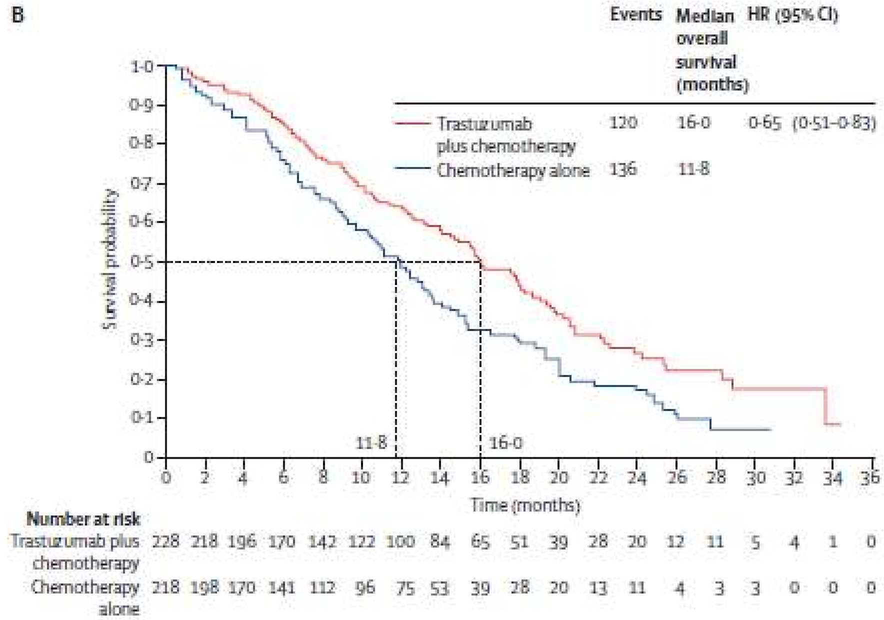
- Supportive care, pain control
- Palliative resection for bleeding
- Palliative bypass (GJ) for outlet obstruction
- Systemic chemotherapy
- Endoscopic stenting

Chemotherapy

- Primary
- Adjuvant
- Neoadjuvant

ToGA trial – HER-2 mutation





Bang YJ, *et al*, Lancet 2010

Endoscopic stenting

Value of diagnostic imaging

- Staging – chest x-ray, CAT scan, PET scan
- Diagnosis
  - Intestinal obstruction
  - Malignant biliary obstruction
  - Malignant ureteric obstruction
- Monitoring response to treatment